Alex Mateescu

IT Service Engineer | Identity and Infrastructure Security

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PROFILE

IT Service Engineer running a hybrid Microsoft environment across two UK sites: Active Directory on Windows Server 2022, Microsoft Entra ID synced through Entra Connect, Microsoft 365 and Exchange Online. Owns the identity lifecycle for 100+ users, patching and vulnerability remediation for about 120 Windows endpoints, and the technical controls behind a first-time Cyber Essentials pass. Outside work, builds labs and tools on the security side of the same stack: Entra ID governance, Windows privilege escalation and its detection in Splunk, and endpoint posture monitoring in PowerShell. Working towards an identity-focused Infrastructure Security Engineer role.

TECHNICAL SKILLS

- **Identity and Access:** Microsoft Entra ID, Active Directory Domain Services (AD DS), Entra Connect (hybrid identity), Conditional Access, multi-factor authentication (MFA), Privileged Identity Management (PIM), access reviews, entitlement management, SSPR, SAML SSO, joiner-mover-leaver (JML), Group Policy (GPO), RBAC, least privilege, Zero Trust
- **Security Operations:** vulnerability management (Qualys, OpenVAS), patch management, endpoint detection and response (EDR), Windows hardening, SIEM (Splunk, SPL), Sysmon, Windows event logs, MITRE ATT&CK, Nmap, Cyber Essentials
- **Infrastructure and Cloud:** Windows Server 2022, VMware ESXi, DNS, DHCP, file shares and NTFS permissions, Microsoft 365, Exchange Online, Microsoft Azure, SPF, DKIM, DMARC
- **Networking:** VLAN segmentation, TCP/IP, Wi-Fi, firewall rules, WireGuard VPN, remote access
- **Automation and Endpoint:** PowerShell (Microsoft Graph SDK, Pester), Python, GitHub Actions, ManageEngine Endpoint Central, ManageEngine ADManager Plus, Windows 11, Bitwarden, Snipe-IT

PROFESSIONAL EXPERIENCE

IT Service Engineer (Systems Engineer scope) | Swagelok Central, Manchester, UK | Jul 2025 – Present

- Manage joiners, movers and leavers in Active Directory and Microsoft Entra ID for 100+ users, covering group membership, licences and MFA enrolment.
- Built Group Policy objects that deploy the Qualys Cloud Agent, EDR and ManageEngine agents to every domain-joined device and reinstall any agent that fails a health check.
- Patch about 120 Windows endpoints through ManageEngine Endpoint Central and report compliance against agreed timescales.
- Triage Qualys findings by severity, remediate them and rescan to confirm each fix, so the open count falls with every patch cycle.
- Delivered the technical controls for the 2025 Cyber Essentials certification, passed at the first attempt: access reviews, endpoint policy changes, patch timescales and supporting SOPs.
- Snapshot server VMs on VMware ESXi before patching or configuration changes, keeping a rollback point if an update fails.
- Resolve 1st and 2nd line incidents across Microsoft 365, DNS, DHCP, VLANs, Wi-Fi and remote access.
- Set up Snipe-IT as the asset register for both sites, and traced, labelled and recorded every patch panel port to its wall socket.

Key project: Bristol and Manchester domain consolidation

- Consolidated two site domains into a single Active Directory domain, with a Windows Server 2022 domain controller VM on VMware ESXi in each office (AD DS, DNS, DHCP); redesigned the OU structure and verified replication.
- Migrated 60+ user accounts with ADManager Plus and matched each to its existing Entra ID identity, giving staff one synced password for Windows and Microsoft 365.
- Replaced manual domain joins with a PowerShell script deployed through ManageEngine that moved each Windows 11 laptop into the new domain and its correct OU, then ran each user's cutover: new credentials, Microsoft 365 reconnection, OneDrive data and bookmarks, and Bitwarden onboarding.
- Moved file shares with Robocopy, preserving NTFS permissions.

Shipping Auditor | Amazon, UK | Sep 2018 – Sep 2024

- Audited outbound shipment data and raised discrepancies for correction.
- Diagnosed faults on handheld scanners and warehouse systems with IT and operations staff, and trained new associates to use them.

IT Technician | DUEPUNTOZERO, Italy | Jan 2017 – Aug 2018

- Assembled and deployed Windows desktops from bare hardware: BIOS settings, OS installation, drivers and peripherals.
- Fixed hardware, software, user profile and performance problems for end users.
- Configured user accounts, IP addressing, DNS and LAN connections.

PROJECTS

Write-ups, code and tests at hawkode.github.io *and* github.com/Hawkode

IAM Lab in Microsoft Entra ID, with Entra IAM Automation

- Designed an identity environment from an empty Entra ID tenant: attribute-driven dynamic groups, joiner-mover-leaver tested end to end, MFA and Conditional Access enforced after a report-only phase, legacy authentication blocked, and a break-glass account proven in the sign-in logs.
- Removed standing admin roles with PIM, confirmed by before-and-after role assignments, and configured access reviews, entitlement management and SAML SSO across all four SC-300 domains.
- Automated cloud and hybrid joiners, movers and leavers, access review campaigns and read-only reports on standing admins, stale accounts, MFA registration and risky app consents in PowerShell with the Microsoft Graph SDK, linted in CI.

Windows Endpoint Security: Attack, Detect, Harden

- Escalated from a low-privilege foothold to SYSTEM on a Windows lab VM through service misconfigurations, SeImpersonatePrivilege token abuse and SAM/SYSTEM hive theft, recording the Security log and Sysmon events each step leaves.
- Wrote Splunk SPL searches and scheduled alerts for those techniques, mapped to MITRE ATT&CK T1134, T1003.002, T1543.003 and T1548.
- Rebuilt Windows Health Monitor as a PowerShell module that checks Defender, firewall, patch age, BitLocker and log clearing, sends findings and a heartbeat to the event log for Splunk, and refuses to run as SYSTEM from a user-writable folder; 50 Pester tests run in CI on PowerShell 5.1 and 7.

MailGuard: phishing triage CLI (Python)

- Developed a tool that scores suspicious .eml files on Microsoft 365 verdict headers, SPF, DKIM and DMARC alignment, unwrapped Safe Links, attachments and sender identity, without opening a link or running an attachment.

CERTIFICATIONS

- CompTIA Security+, Network+ and A+ (Secure Infrastructure Specialist)
- Cisco Certified Network Associate (CCNA)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900)

In progress

- Microsoft Certified: Identity and Access Administrator Associate (SC-300), target 2027
- Microsoft Certified: Security Operations Analyst Associate (SC-200), target 2027
- CompTIA Cybersecurity Analyst (CySA+), target 2027
- TryHackMe Pre Security (SEC0) and Cyber Security 101 (SEC1), target Dec 2026

EDUCATION

- BSc (Hons) Computer Science | Arden University | 2024 – 2025
- HND Digital Technology | GBS | 2021 – 2023